

# SAT.O–Blockwave: Unified Action — Corrected, Dimensionally Consistent Formulation

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## Abstract

A single, dimensionally consistent formulation of the SAT.O/Blockwave unified-action model. This source consolidates the full field content, specifies SI-consistent units for all quantities, presents the corrected Lagrangian density, and lists the resulting coupled partial differential equations. An appendix provides line-by-line dimensional bookkeeping.

## Conventions and SI base units

We use SI base symbols for dimensional analysis:  $M$  (mass),  $L$  (length),  $T$  (time). The metric signature is not required for this non-relativistic morphogenic presentation.

## 1 Field definitions and identities

We define the basic fields used throughout:

$$\mathbf{v}(\mathbf{r}, t) = \frac{\mathbf{J}(\mathbf{r}, t)}{\rho(\mathbf{r}, t)}, \quad [\mathbf{v}] = LT^{-1}, \quad (1)$$

$$S[\Phi] \equiv (\nabla\Phi), \quad [\nabla\Phi] = LT^{-2}. \quad (2)$$

The curvature tensor for the scalar field is the rank-2 morphotensor  $\mathbf{C} = \nabla \otimes \nabla\Phi$ .

| Symbol                   | Physical meaning and SI dimensions                                                             |
|--------------------------|------------------------------------------------------------------------------------------------|
| $\rho$                   | mass (energy) density, $[\rho] = ML^{-3}$                                                      |
| $\mathbf{J}$             | mass flux density, $[\mathbf{J}] = ML^{-2}T^{-1}$                                              |
| $\mathbf{v}$             | velocity field, $[\mathbf{v}] = LT^{-1}$                                                       |
| $\Phi$                   | morphogenic potential per unit mass, $[\Phi] = L^2T^{-2}$                                      |
| $W(\Phi)$                | potential per unit mass (same units as $\Phi$ ), $[W] = L^2T^{-2}$                             |
| $U(\Phi) = \rho W(\Phi)$ | energy density, $[U] = ML^{-1}T^{-2}$                                                          |
| $\kappa$                 | curvature coupling, $[\kappa] = ML^{-3}T^2$                                                    |
| $c$                      | characteristic wave speed, $[c] = LT^{-1}$                                                     |
| $\gamma$                 | damping, $[\gamma] = T^{-1}$                                                                   |
| $\lambda$                | dimensionless coupling (appears in Lagrangian as $\lambda \rho \Phi \nabla \cdot \mathbf{v}$ ) |
| $P$                      | pressure, $[P] = ML^{-1}T^{-2}$                                                                |

Table 1: Primary symbols and their SI-base dimensions used in the text.

## 2 Corrected Lagrangian density

We choose a dimensionally consistent Lagrangian density (energy density units):

$$\mathcal{L} = \frac{1}{2}\rho \mathbf{v}^2 - \rho W(\Phi) - \frac{\kappa}{2}(\nabla \Phi)^2 + \lambda \rho \Phi (\nabla \cdot \mathbf{v}). \quad (3)$$

Here  $W(\Phi)$  is the potential per unit mass; the coupling  $\lambda$  is taken dimensionless and  $\rho$  supplies appropriate units in the interaction term.

## 3 Euler–Lagrange system (resulting PDEs)

Variation of  $\mathcal{L}$  with respect to the hydrodynamic variables and  $\Phi$  yields the following coupled PDE system.

### Mass continuity

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0. \quad (4)$$

### Momentum balance

The momentum equation (obtained from  $\delta \mathcal{L} / \delta \mathbf{v}$ ) reads:

$$\rho (\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}) = -\nabla P - \rho \nabla W(\Phi) - \kappa (\nabla^2 \Phi) \nabla \Phi + \lambda \nabla (\rho \Phi), \quad (5)$$

where the curvature-derived body force has the form  $\mathbf{f}_\kappa = -\kappa(\nabla^2\Phi) \nabla\Phi$  (see Appendix for derivation). The last term arises from the interaction piece and is dimensionally consistent when  $\lambda$  is dimensionless.

## Scalar (per-unit-mass) evolution

The scalar evolution equation (variation w.r.t.  $\Phi$ ) in mass-normalized form is:

$$\partial_t^2\Phi + \gamma \partial_t\Phi = c^2 \nabla^2\Phi - \frac{dW}{d\Phi} + \frac{\lambda}{\rho} \nabla \cdot (\rho\mathbf{v}). \quad (6)$$

Here  $dW/d\Phi$  carries units of  $T^{-2}$  so that each term has the same dimensions  $[\Phi]/T^2 = L^2T^{-4}$ .

## 4 Energy and conserved quantity

Define total energy (for  $\gamma = 0$ ) as

$$E_{\text{tot}} = \int_V \left[ \frac{1}{2}\rho\mathbf{v}^2 + \rho W(\Phi) + \frac{\kappa}{2}(\nabla\Phi)^2 \right] dV. \quad (7)$$

Direct computation, using the PDEs above, shows that  $dE_{\text{tot}}/dt = 0$  when  $\gamma = 0$  and appropriate boundary conditions apply (see Appendix).

## Dimensionless rescaling

Introduce characteristic scales  $L_0$ ,  $T_0$ ,  $\rho_0$ , and  $\Phi_0$ . Define nondimensional variables

$$\tilde{\mathbf{r}} = \frac{\mathbf{r}}{L_0}, \quad \tilde{t} = \frac{t}{T_0}, \quad \tilde{\rho} = \frac{\rho}{\rho_0}, \quad \tilde{\Phi} = \frac{\Phi}{\Phi_0}, \quad \tilde{\mathbf{v}} = \frac{\mathbf{v}T_0}{L_0}.$$

The governing equations can be written in terms of dimensionless control numbers such as

$$\text{Re}_\Phi = \frac{\rho_0 L_0^2}{\kappa T_0^2}, \quad \text{Ma}_\Phi = \frac{V_0}{c}, \quad \text{Da}_\Phi = \frac{W(\Phi_0)T_0^2}{\Phi_0}.$$

# Appendix: Dimensional bookkeeping and derivations

## A. Units for the curvature force

The curvature energy density is  $\frac{\kappa}{2}(\nabla\Phi)^2$ . Variation with respect to  $\Phi$  yields a functional derivative that produces a body-force density of the form

$$\mathbf{f}_\kappa = -\kappa (\nabla^2\Phi) \nabla\Phi + (\text{higher-order gradient terms}).$$

Check units:  $[\nabla^2\Phi] = T^{-2}$ ,  $[\nabla\Phi] = LT^{-2}$ ,  $[\kappa] = ML^{-3}T^2$  so  $[\mathbf{f}_\kappa] = ML^{-2}T^{-2}$  as required for a force density.

## B. Euler–Lagrange to PDE mapping

A short derivation of the scalar equation: compute  $\delta\mathcal{L}/\delta\Phi$  and apply continuity where necessary; one finds the mass-normalized wave-like equation shown in Eq. (6).

## C. Energy conservation sketch

Multiply Eq. (5) by  $\mathbf{v}$  and integrate, combine with Eq. (6) multiplied by  $\rho \partial_t\Phi$ , and integrate by parts. Under vanishing-flux boundary conditions the total energy time derivative vanishes for  $\gamma = 0$ .